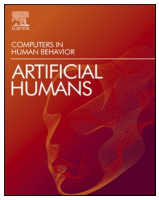
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Optimizing human-AI collaboration: Effects of motivation and accuracy information in AI-supported decision-making

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ABSTRACT

Artificial intelligence (AI) systems increasingly support human decision-making in fields like medicine, management, and finance. However, such human-AI (HAI) collaboration is often less effective than AI systems alone. Moreover, efforts to make AI recommendations more transparent have failed to improve the decision quality of HAI collaborations. Based on dual process theories of cognition, we hypothesized that suboptimal HAI collaboration is partly due to heuristic information processing of humans, creating a trust imbalance towards the AI system. In an online experiment with 337 participants, we investigated motivation and accuracy information as potential factors to induce more deliberate elaboration of AI recommendations, and thus improve HAI collaboration. Participants worked on a simulated personnel selection task and received recommendations from a simulated AI system. Participants' motivation was varied through gamification, and accuracy information through additional information from the AI system. Results indicate that both motivation and accuracy information positively influenced HAI performance, but in different ways. While high motivation primarily increased humans' use in high-quality recommendations only, accuracy information improved both the use of low- and high-quality recommendations. However, a combination of high motivation and accuracy information did not yield additional improvement of HAI performance.

1. Introduction

Artificial intelligence (AI) systems are increasingly implemented to improve decision-making in multiple domains, such as management, medicine, and finance (Iyer, 2021; Lange & Pfarrer, 2017; Langer et al., 2021; Milana & Ashta, 2021; Reddy et al., 2020). Although the construct of artificial intelligence, or "AI", is still discussed somewhat controversially, AI-based decision-making systems can be understood as computer systems that autonomously learn patterns and rules on a data basis, and apply them to specific decision-making tasks (Samoili et al., 2020). AI decision-making systems are characterised by a high degree of accuracy in their predictions that often exceeds the decision-making quality of humans (Green & Chen, 2019; Jacobs et al., 2021). However, in areas where a decision directly affects humans (e.g., in personnel selection), legislation prohibits fully automated decisions based on AI systems. For instance, AI-generated decision rules are often too complex to be comprehensible by humans, which makes it difficult to control AI systems' suggestions, for instance, with respect to potential discrimination of minorities. As a consequence, countries around the world (e.g., in the US and the European Union) have refined the global regularisation of AI use in recent years, often requiring humans to remain in control of AI-based decision contexts (e.g., Chae, 2020).

Such human-AI (HAI) collaboration produces new problems, since research suggests that humans often use AI predictions inadequately and tend to make worse decisions than AI systems would have made in isolation (Bansal et al., 2021; Zerilli et al., 2022). Such suboptimal usage has been explained both by users' over-trust or "algorithm appreciation" in AI systems (i.e., uncritical acceptance of AI systems' recommendations at first contact, e.g., Logg et al., 2019) and under-trust or "algorithm aversion" in AI systems (i.e., exaggerated aversion towards AI systems' recommendation after observing the system making an error, e.g, Burton et al., 2020; Dietvorst et al., 2015), depending on the circumstances (see also Hertel et al., 2023). While many studies either focus on over-trust or under-trust, it seems to be important to achieve a balance of both adequate trust and sceptical usage of AI systems' recommendations when appropriate, generally described as "well--calibrated trust" (Zerilli et al., 2022; Zhang et al., 2020). However, such calibration of trust in AI systems is not trivial given the complexity of

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AI-based processes.

One potential way to achieve well-calibrated trust discussed by both researchers and practitioners are explainable AI (XAI) approaches. XAI approaches augment AI system outputs with explanations on system decision processes or regarding single system outputs. For example, so-called "local explanations" XAI approaches can show which features most strongly impact a system's recommendation. Thereby, XAI can make system processes or outputs more comprehensible for users, so that users can to tell when to rely and/or when to reject a system output (Langer et al., 2022). While initial research suggests that such XAI approaches only slightly improve the decision quality of HAI collaborations (Abdul et al., 2020; Ngo & Krämer, 2022), we reason that it is necessary to better understand motivational and cognitive processes that may foster or hinder the effects of XAI approaches. Indeed, if the additional information provided by XAI approaches is not appropriately processed by human users, their potential to improve joint decision-making might be wasted.

We use dual process theories of cognition (e.g., Chaiken, 1980; Strack & Deutsch, 2004) to predict that XAI approaches only lead to appropriate trust if they are processed systematically (instead of heuristically). Basically, dual process theories differentiate between two different modes of cognitive information processing, for instance, during decision making. One is a systematic mode, consisting of careful and deliberative processing of information. The other is a heuristic mode, relying on simplified decision rules ("rules of thumb") or 'heuristics' that are stored in memory and generalized across problems. While the former is more likely to lead to high quality decisions, the latter is quick and can safe considerable cognitive resources. We propose that high motivation to perform well in a HAI task is crucial for humans to deliberately process AI recommendations and explanations. Moreover, we propose that accuracy information[1] (information from the AI system about the accuracy of its recommendation) can help users to calibrate their trust in the AI system, providing indicators of which AI explanations might be questioned, and when users' trust in the AI system may or may not be appropriate. We examined our propositions in an online experiment simulating a personnel selection context. In so doing, we focussed on an intermediate level of automation (e.g., Kaber & Endsley, 2004), where the AI system acquires and analyses information and recommends a decision, which the human partners can then examine and consider as part of their decision. The relationship between human and AI is seen, as common in recent literature, as "teamwork", with AI occupying legitimate roles in work tasks and communicating with human team members, bringing new facets to the concept of human-computer interaction (O'Neill et al., 2023a; 2023b). In addition to effects of motivation and accuracy information on HAI decision quality, we also explored the influences of motivation and accuracy information on users' trust, using both self-reports and behavior indicators.

This research makes the following contributions to the literature on AI-based decision-making: Instead of focusing on only one potential influence to improve HAI collaboration, we considered both the influence of motivation and effects of accuracy information on decision-making quality, enabling us to also examine potential interactions between those factors. Moreover, we explored both the dangers of over-trust and under-trust towards AI systems considering appropriate trust calibration. In doing so, we specifically examined the beneficial effect of motivation and accuracy information on the usage of high- and low-quality explanations. Together, this research advances the theoretical understanding of decision-making processes of users' collaboration with AI systems.

[1] We use the term "accuracy information" instead of "confidence information" to emphasize that the information refers to the performance of the system and not to observers' beliefs.

2. Related work

2.1. Joint Human-AI decision-making

Recent literature has investigated whether human and AI system decision-making can be combined to harvest the best from both worlds (Bansal et al., 2021). Tasks in which humans perform well (e.g., intuitive decision-making on very little information) often differ from tasks in which machines perform well (e.g., tasks requiring complex algorithmic calculation). A combination of human and AI systems should intuitively result in complementary performance (i.e., better than each alone) since they could combine their strengths and compensate for each other's weaknesses (Mosier & Manzey, 2019). AI can handle complex and unambiguous decisions efficiently, while humans are able to intuitively decide in edge cases and include ethical considerations.

However, research to date suggests that while human-AI collaborations sometimes perform better than humans alone, their performance usually falls short of an AI's solo performance. This has been shown for a variety of tasks, including selecting clinical treatments (Jacobs et al., 2021), predicting clinical emergencies during operations (Lundberg et al., 2018), detecting fake hotel reviews (Lai & Tan, 2019), predicting loan defaults (Zhang et al., 2020), and estimating house prices (Poursabzi-Sangdeh et al., 2018). It seems as if human-AI collaborations have difficulties in realising their full potential.

In light of these results, one might consider to remove humans entirely from HAI decision-making, especially in high-risk contexts where false decisions have dire consequences. However, automation without supervision can be also dangerous. A classic example is the hiring algorithm Amazon.com employed in 2014, which had been trained to preselect applications based on 10 years of prior hiring decisions. In 2015, the engineers realized that due to high dominance of male hires in the past, the algorithm taught itself to systematically rate women's applications as inferior to men's, even though the sex of the applicants was not explicitly fed into the system (Cook, 2018). To avoid such pitfalls of automation in high-risk contexts, such as personnel selection, the General Data Protection Regulation prohibits fully automated decision-making and calls for humans to be involved in decision-making. Current proposals for legislation around the globe are even more explicit and call for human oversight when AI system are used in high-risk domains (Green, 2022). For instance, the European Commission lists human oversight as one of its key requirements in its proposal for a Europe-wide AI Act (European Commission, 2021; see also Chae, 2020, for similar proposals in the US), and the German Trade Union Confederation considers joint decision-making with humans to be a control mechanism necessary in personnel selection and urges more transparency in AI systems (DGB, 2022). Therefore, joint human AI decision-making will remain the most likely way of implementing AI systems in decision processes in the future, and improvement of their joint decision-making abilities is needed.

2.2. Potential ways to improve Human-AI decision-making

A prominent problem in the use of AI systems is the "black box" problem: How can we effectively work together with something we cannot comprehend? Especially in recent years, research has increasingly developed methods that make AI algorithms more comprehensible (Adadi & Berrada, 2018). Research on "explainable AI", or XAI, assumes that explanations regarding systems' processes and outputs allow for better human control and usage of AI systems (Arrieta et al., 2020). This emphasis on explanations is even embedded in European law, with the GDPR granting humans that were made subject to algorithmic decisions a "right of explanation" (Hunkenschroer & Luetge, 2022).

A common approach in XAI is to offer local explanations, where the goal is to explain individual decisions (as opposed to global explanations, giving information about general decision rules) (Speith, 2022). For instance, feature importance methods allow users to gain insight

into what information was deemed most relevant to a specific decision of the system. If an algorithmic system diagnoses a risk for hypoxemia (a dangerously low level of oxygen in the blood) during an operation (Lundberg et al., 2018), it could justify this decision by highlighting important features of the patient that have led to this diagnosis. The human in charge of the final diagnoses and treatment decisions can use this information for either confirmation (e.g., high breathing rate would be a valid risk factor for hypoxemia) or dismiss it (e.g., high body temperature would not be a valid risk factor for hypoxemia).

However, employing feature importance (or other XAI methods) has not solved the problem of suboptimal performance in AI-supported decision contexts. Specifically, providing the commonly used local explanations does not necessarily lead to better decisions (Jacobs et al., 2021; Ngo & Krämer, 2022; Zhang et al., 2020). For example, Jacobs et al. (2021) showed that clinicians did not improve their treatment selection accuracy when they received a correct AI system recommendation paired with a list of the most important features leading to the recommendation compared to when they made the decision independently, without a support system. If the recommendation was wrong, the explanation even decreased the clinician's accuracy in comparison to deciding without a system.

2.3. Suboptimal decision-making due to miscalibration of trust

But why do XAI approaches not automatically lead to better performance? The answer potentially lies in the impact of XAI on trust. A key motivation of XAI is to simultaneously increase users' trust into a trustworthy AI and users' distrust into a non-trustworthy AI (Jacovi et al., 2021). Trust is defined as a "willingness to make themselves vulnerable" (Mayer et al., 1995), or, more specifically in the case of AI decision aids, as "the willingness of a person to be vulnerable to the actions of an AI decision aid, based on the expectation that it will perform a decision-making task important to the trustor" (Solberg et al., 2022). Since the recommendations of AI systems are (even with explanations) not fully comprehensible, decision-makers are vulnerable to false recommendations and have to trust in the AI system's capabilities to use them (Buçinca et al., 2021; Hertel et al., 2023).

However, instead of accurately calibrating trust, explanations can also lead to inadequately high and low trust, depending on the circumstances. Dual process theories of cognition (Chaiken, 1980; Strack & Deutsch, 2004) suggests that during decision-making, humans rely primarily on fast and efficient thinking, known as heuristic thinking. This type of thinking reduces effort and is sufficient for everyday decisions. However, it can lead to cognitive biases and simplified decisions (generalized "rules of thumb"). Systematic thinking is based on more deliberate information processing and produces more considered decisions, but is more effortful and used less frequently. Heuristic decision-making is what possibly impedes the positive effect of explanations in HAI decision tasks. For example, offering explanations per se can lead to an increase in trust in the system's recommendations, independent of the quality of the explanation (Chromik et al., 2021). This may be due to heuristic decision-making, where people take the presence of an explanation as an indication of trustworthiness without systematically evaluating the content of the explanations (Buçinca et al., 2021). This can even increase low-quality explanations and hamper the calibration of users' trust (Bansal et al., 2021; Buçinca et al., 2021; Chromik et al., 2021; Eiband et al., 2019).

While over-trust is considered as more common problem (Zerilli et al., 2022), heuristic decision-making can also lead to unwarranted low trust in an AI system. For example, if an explanation is poorly designed and requires high cognitive effort, it can reduce users' willingness to engage with the explanation and create unjustified low trust (Kaur et al., 2020; Kulesza et al., 2013).

The key to improve decision-making quality potentially lies in facilitating the calibration of trust, i.e., to use recommendations with the help of explanations in accordance with their quality (McGuirl & Sarter, 2006; Zerilli et al., 2022; Zhang et al., 2020). To achieve this improvement, it appears promising to explore ways to facilitate deliberate decision-making of humans in AI-supported decision tasks (Buçinca et al., 2021; Kaur et al., 2020).

2.3.1. Motivation as factor for trust calibration

The first promising way to facilitate systematic decision-making, and thus to facilitate trust calibration through explanations in HAI tasks is to increase people's motivation to perform well (e.g., Chaiken, 1980). Such motivation can be influenced by extrinsic factors (e.g., triggered by rewards, benefits or sympathies associated with a certain given behavior) or intrinsic factors (e.g., by one's own preferences, personality, and goals). Previous research has shown that people report a low desire to understand a system, especially if the stakes of the supported decision are not perceived as high (Bunt et al., 2012). The effort to understand a system may subjectively outweigh the perceived benefits (Gregor & Benbasat, 1999). This can lead to either dismissing good recommendations if they are not in line with one's intuitions or accept them uncritically to reduce effort. Research has shown that forcing participants to elaborate explanations more deliberately resulted in better joint decision-making with less over-trust, but only for participants with low intrinsic motivation to exert cognitive effort, (Buçinca et al., 2021), also known as "need for cognition" (Cacioppo & Petty, 1982). This improvement in decision-making suggests that explanations can indeed improve HAI team decision-making, but only if the explanations are evaluated deliberately and not heuristically. However, participants who were forced to engage more critically with the explanation reported lower satisfaction with the system despite their improved performance, making it less attractive as an option for HAI task design. Since the authors only found a beneficial forcing effect for participants with a low need for cognition, it seems that people with high intrinsic motivation to exert cognitive effort already deliberately process the information (Buçinca et al., 2021). We conclude that sufficient motivation is a crucial factor for people to deliberately interact with and benefit from XAI explanations. We expect participants with high extrinsic motivation to also exert cognitive effort more willingly and deliberately process the explanation of an AI system's recommendation. This may allow for trust that is better calibrated (trusting trustworthy recommendations and not trusting untrustworthy recommendations) and lead to better performance in HAI decision tasks. Thus, we propose *H1.1 Extrinsic motivation positively affects performance of human- AI collaborations in decision tasks with explanations.*

However, humans' motivation to elaborate AI recommendations may not only come from extrinsic influences. Users with a high need for cognition may be less strongly motivated by extrinsic incentives because they are already intrinsically motivated in cognitive tasks. Thus, we propose *H1.2: The positive effect of extrinsic motivation on performance of human-AI collaborations in decision tasks with explanations is moderated by users' need for cognition, so that high need for cognition reduces the effect of extrinsic motivation.*

If motivated participants exert more effort in joint decision tasks, this should also lead to better detection of low-quality recommendations, i.e., "mistakes" by the system, and reduce the prevalent problem of over-trust. This may also lead to a decrease the trust that motivated people have in the system, as suggested by previous research showing that people lose more trust in system recommendations when they see those systems err, compared to when a human errs (Dietvorst et al., 2015). The more deliberate processing of explanations through a higher motivation to perform well results in a decrease of trust in the system, through a higher probability of finding errors in the system's explanations and a correction of heuristic over-trust. However, it is equally conceivable that if the heuristic impulse of a person is to not trust the system, the more deliberate elaboration of the explanation through motivation might as well correct the trust level upwards. Given that the direct effects of motivation on trust are not yet conclusive, we consider the research question *RQ1: How does high extrinsic motivation to perform well in*

human-AI decision tasks influence trust towards the AI system?

2.3.2. Accuracy information as factor for trust calibration

Another promising way to facilitate trust calibration through explanations in HAI tasks is to provide accuracy information. Accuracy information is an estimation of the AI system about the quality of its own recommendation (high/low probability of error) (Zhang et al., 2020). A classification model could, for example, indicate a high probability of accuracy, when the classified case is very common and similar to the data with which the model was trained on. While accuracy information arguably qualifies as an XAI method, it differs from other XAI explanations because it does not provide direct information about the reasoning behind a recommendation, but rather gives meta-information about the quality of the recommendation. Initial studies suggest that providing accuracy information about the AI system's recommendation does indeed improve the quality of HAI decision-making (McGuirl & Sarter, 2006; Zhang et al., 2020). Accuracy information may benefit trust calibration because it can help users to decide when (costly) systematic information processing is needed, i.e., when a recommendation of the AI system has only low accuracy. Therefore, accuracy information can reduce the probability of over-trust. Also, accuracy information may reduce users' cognitive load because the user can concentrate their cognitive resources on recommendations with low accuracy, counteracting the negative effects of high cognitive load on XAI explanation usage (Kaur et al., 2020; Kulesza et al., 2013).

Since accuracy information has the potential to reduce the cognitive effort necessary to use AI systems' explanations and facilitates trust calibration, we propose *H2: Accuracy information positively affects performance of human-AI collaborations in decision tasks with explanations.*

While we argue that accuracy information can reduce effects of over-trust by encouraging deliberate elaboration of AI system recommendations with low accuracy, it is also possible that generally sceptical users feel encouraged to trust an AI system more when the AI system indicates high accuracy of its recommendation. In addition, the low cognitive load associated with heuristic decision making might also have a positive effect on trust in the AI system. Thus, given that the direct effects of accuracy information on users' trust in the AI system are not conclusive, we consider the research question *RQ2: How does accuracy information in human-AI decision tasks influence trust towards the AI system?*

Finally, we expect the positive effects of motivation and accuracy information to be incremental to each other. Since motivation is expected to increase the willingness to exert cognitive effort, and accuracy information provides additional information about the quality of each recommendation, we expect motivated individuals to benefit more from the accuracy information. To test these assumptions, we formulate hypothesis *H3: We expect motivation and accuracy information to have independent effects on performance of human-AI collaborations in decision tasks with explanations, so that the combination of both factors should lead to a better performance than either factor alone.*

3. Method and analysis

3.1. Participants

We gathered a sample of 337 working adults (137 male, 198 female, 2 diverse, $M_{age} = 55.17$ years, age range = 20–85 years) using a non-commercial German online panel. Participants reported little experience with personnel selection (*mean* = 2.3 on a scale from 1 [never engaged with it] to 4 [often engaged with it]) and even less experience with AI systems (*mean* = 1.73 on a scale from 1 [never engaged with it] to 4 [often engaged with it]). Participants were incentivized by access to an exclusive article about AI use in personnel selection.

We excluded participants if they met one of the following criteria: They indicated that their data should not be used, they did not complete the survey, or they showed monotonic responses in consecutive questions even though this led to inconsistent and contradictory answers, indicating careless and thus non-reliable answering.

3.2. Procedure

After starting the online survey, participants responded to items assessing demographic and potentially moderating variables such as *need for cognition* and *affinity for technology*. Next, participants were instructed to imagine being in the following scenario: They are responsible for personnel selection in a company. The company uses software based on artificial intelligence that is supposed to support the selection of suitable applicants. The AI was trained on previous staffing decisions. Participants were instructed to rate the qualification of multiple applicants (see Appendix A.1 for an example of an applicant profile) on a scale between 1 (not qualified at all) to 100 (perfectly qualified) given a set of rules in a job profile (see Appendix A.2). To ensure understanding, participants completed a tutorial where they rated three applicants and were provided with the correct answer in the end of each trial. After the tutorial, the main trials began, where each decision was supported by the AI system that recommended a numeric rating for each applicant on a scale between 1 and 100. This recommendation was, most of the time, close to the optimal applicant rating (for details on the applicant rating task, see section "Performance Task"). In rounds 3 and 6, the recommendation deviated from the optimal decision to simulate a low-quality AI recommendation. To help participants to evaluate the AI recommendations, we included an explanation showing which of the applicant attributes was most influential regarding the final scoring of the system. This realized a feature-relevance explanation approach, a common approach in the field of XAI (Arrieta et al., 2020).

The experiment followed a 2 (motivation through gamification: yes vs. no) x 2 (accuracy information: yes vs. no) between-participants design with one additional within-participant factor (quality of AI recommendation: high vs. low). Thus, a quarter of participants were motivated with gamification, a quarter were provided with accurate information about the AI, a quarter were both motivated and provided with information, and a quarter experienced neither. All participants received an AI system's recommendation and an accompanying explanation. Participants in the control condition received no additional information or motivation. Participants in the motivation condition were (before conducting the tutorial of the performance task and again after finishing it) informed that there will be a leader board at the end of the study indicating which participants showed the best performance. Participants in the accuracy information condition received additional warnings that indicated when the AI system has low certainty that its recommendation is accurate. This accuracy information was designed in a way that the low certainty warnings appeared as text in red under the recommendation ("the AI is rather uncertain about the recommendation") in rounds with low-quality AI system recommendations and in green in rounds with high-quality recommendations ("the AI is rather confident about the recommendation"). The final experimental condition was a combination of the motivation and the accuracy information condition. After performing the task, we captured self-reported trust in the AI system as well as participants' motivation to enable a manipulation check.

3.3. Performance task

We measured participants' performance in the simulated personnel selection task where they had to rate multiple applicants' suitability for a job based on a job profile (see Appendix A.1). The participants had to rate every profile on a scale from 1 (not qualified at all) to 100 (perfectly qualified). To assess the qualification for this hypothetical position, they received a set of rules regarding how to evaluate the profiles (see Appendix A.2). These rules were the same rules underlying our formula to calculate the optimal rating for applicants, but instead of presenting the formula, we paraphrased it without concrete values (for exact wording,

see Appendix A.2). The formula to calculate the "optimal score" is as follows: There are three test dimensions, and each dimension consists of two tests whose scores are averaged, resulting in three scores. These scores fall between 1 and 100; test one counts in with a factor of 3, test two with a factor of 0.7, and test three with a factor of 2.3. Then, if the work experience is 1, 25% of the resulting score is subtracted if certificate A is missing and/or by 20% if certificate B is missing. Finally, if the work experience is 2 rather than 3, then the value is reduced by 5% (see Appendix A.3 for the code to calculate the optimal score). Participants received no information about the rule-based nature of the algorithm, and were informed that the system had been trained on previous staffing decisions.

The system provided an applicant rating that in rounds 1,2,4, and 5 showed a value with a deviation of 5 points below or above the correct score. Whether the deviation from the score was above or below the optimal value was consistent for all participants. In the remaining two rounds, the AI system recommended a value 20 points below or above the correct score. Each recommendation of the system came with an explanation that was designed to enable participants to recognize the two faulty recommendations. Specifically, the explanation contradicted the evaluation rules. In round 3 the recommendation of the system deviated 20 points from the optimal value, with the explanation that the second tests were the most strongly weighted factor for its recommendation; the rules, however, state that the second tests should be weighted least strongly. In round 6, the recommendation deviated because of the applicant's work experience, which, according to the rules, should have a weaker impact if work experience is a 2 out of 3.

3.4. Measures

Self-reported trust was measured after the performance task with the single item "I perceived the AI as trustworthy" on a scale from 1 (strongly disagree) to 7 (strongly agree). Need for cognition was measured with a six-item short scale by Coelho et al. (Lins de Holanda Coelho et al., 2020) translated into German by Bless et al. (1994). Participants answered on a scale from 1 (does not apply to me at all) to 7 (fully applies to me). A sample item for this scale was "I would prefer complex to simple problems". Affinity for Technology-Interaction (ATI) was measured with a German version of the scale (Franke et al., 2019). Participants answered on a scale from 1 (is not true at all) to 6 (is completely true). A sample item for this scale was "I like to occupy myself in greater detail with technical systems".

We calculated the performance score of each participant by determining the deviation of their rating from the optimal score. Depending on the analysis, these scores were averaged over all trials for between-participant analyses or considered individually for each trial to allow for within-participant analysis. Please note the following when interpreting results regarding the participants' performance: If a participant always exactly followed the AI-based recommendation, they would reach an average of 10 points of deviation from the correct score. This can be considered the baseline performance, with values under 10 indicating successful joint decision-making: decision-making that is better than always following the system-based advice.

We additionally calculated the variable *deviation from AI recommendation* by determining the average deviation of their rating from the recommended score. To be clear, applicant performance was defined as the deviation from the 10-point deviation from the optimal score that participants would have if they always followed the system recommendation. *Deviation from AI recommendation* is a measure of how strongly participants deviated from the AI-based recommendation.

3.5. Manipulation check

To examine whether our experimental manipulation of participants' motivation was successful, i.e., whether participants were more motivated in the two conditions with the gamification intervention as compared to the two conditions without gamification, participants were asked to answer the question "I was motivated to perform well at the task" on a scale from 1 (strongly disagree) to 7 (strongly agree).

4. Results

All analyses were conducted using either Python 3.10 or R 3.6. Table 1 provides an overview of descriptive statistics such as correlations, means, and standard deviations of all dependent and independent variables.

4.1. Trust and manipulation check

The self-reported measure of trust in the AI system and the deviation from the AI recommendation (a behavioral indicator of trust, where lower deviations indicate higher trust) correlated negatively, as expected (see Table 1). Moreover, although participants did not report higher motivation to perform well in the conditions with gamification, $F(1,335) = 0.15$, $p = .69$, the gamification intervention did increase their performance (see section 4.3.1).

4.2. Effects of the quality of AI-based recommendations

To examine whether the AI-based recommendation affected participants' performance, we conducted an ANOVA. The results showed that participants performed significantly worse in trials with low-quality recommendations, with a one-way ANOVA resulting in $F(1,336) = 46.80$, $p < .001$, $\eta^2 = 0.067$. Participants also deviated more strongly from low-quality recommendations, $F(1,336) = 41.50$, $p < .001$, $\eta^2 = 0.389$, suggesting that they were able to consider the quality of recommendations made by the AI system.

4.3. Testing the hypotheses

4.3.1. Motivation

To test our hypotheses, we conducted a mixed ANOVA with two between factors (motivation, accuracy information) and one within factor (the different trials) and performance as a dependent variable to consider between and within factors equally. H1.1 stated that *extrinsic motivation positively affects performance of human-AI collaborations in decision tasks with explanations.* Indeed, the conditions with high motivation came closer to the right answer than participants with low motivation, with $F(1,333) = 4.43$, $p <. 01$, $\eta^2 = 0.003$ (see Table 3 for mean differences between high- and low-motivation groups; see Table 4 for ANOVA results) as expected in H1.1. To further investigate this result, we examined in what rounds participants showed stronger performance. The results indicated an interaction effect between quality of recommendation and the effect of motivation on performance, $F(1,333) = 40.83$, $p <. 01$, $\eta^2 = 0.06$. Participants' performance more strongly improved in rounds where the AI system gave a high-quality recommendation than when it gave a low-quality recommendation.

H1.2 stated that *the positive effect of extrinsic motivation on performance of human-AI collaborations in decision tasks with explanations is moderated by users' need for cognition, so that high need for cognition reduces the effect of extrinsic motivation.* However, need for cognition did not moderate the effect of motivation on the performance with $F(1,335) = 0.05$, $p = .82$. Notably, the average need for cognition score was very high in the sample ($M = 5.18$, $SD = 0.89$ on a seven-point scale), indicating that most participants were intrinsically motivated to exert cognitive effort (see Table 1).

In RQ1, we wanted to explore *how high extrinsic motivation to perform well in human-AI collaborations in decision tasks with explanations influences trust towards the AI system.* We explored this research question by first testing differences in self-reported trust between the high- and low-motivation conditions. The ANOVA showed no significant effect of motivation on self-reported trust $F(1,335) = 3.63$, $p = .05$. Since prior

Table 1
Means, standard deviations, and correlations for all constructs.

Variable	*M*	*SD*	1.	2.	3.	4.	5.	6.	7.	8.
1. Need for Cognition	5.18	0.89	*.79 (.75-.82)*							
2. Affinity for Technology	3.99	1.03	.32*	*.93 (.92-.94)*						
3. Experience in Personnel Selection	2.30	1.05	.22*	.03	–					
4. Experience with AI	1.73	0.93	.16*	.16*	.23*	-				
5. Trust	4.07	1.21	.01	.09	-.03	-.03	–			
6. Motivation	5.45	1.25	.06	.06	-.05	.01	.24*	–		
7. Deviation from correct score (Performance)	8.63	3.90	-.03	-.05	.10	.09	.01	-.08	–	
8. Deviation from AI Recommendation	10.6	4.26	.00	.00	.00	.07	-.30*	-.02	.38*	–

Note. Need for cognition, affinity for technology, trust and motivation were measured on a seven-point scale with 1 indicating a low and 7 a high score. Experience with personnel selection and artificial intelligence were measured on a scale from 1 (little to no experience) to 4 (a lot of experience). Performance and deviation from AI recommendation ranges from 0 to 70, with a higher value indicating more deviation from the correct value (performance) or the AI recommendation (deviation from AI-based recommendation). Reliability of multi-item scales can be found in italics with 95% confidence intervals. *$p < .05$.
$N = 337$.

Table 2
Performance means and standard deviations divided by trials with high- and low-quality AI recommendations.

Condition	Performance: Low-Quality Recommendation		Performance: High-Quality Recommendation		Deviation from AI: Low-Quality Recommendation		Deviation from AI: High-Quality Recommendation	
	M	SD	M	SD	M	SD	M	SD
Motivated	10.06	5.72	6.90	3.17	16.52	6.95	6.74	3.44
Not Motivated	9.80	5.04	8.14	4.25	16.34	6.96	8.13	4.25
Accuracy information	9.82	5.27	7.02	3.47	16.97	7.09	6.88	3.87
No Accuracy information	10.06	5.54	8.06	4.06	15.82	6.74	8.05	3.89
Overall	9.94	5.39	7.51	3.79	16.43	6.94	7.43	3.92

Note. "Low-quality recommendation" means that the system recommendation deviated by 20 points from the correct value, "high-quality recommendations" means that the system recommendation deviated by 5 points. Low values indicate a good performance, or a low deviation from the value the system recommended. $N = 337$.

Table 3
Performance mean and standard deviation for each trial grouped by high/low motivation and accuracy information.

	T1	T2	T3	T4	T5	T6
	M (SD)	*M (SD)*	*M (SD)*	*M (SD)*	*M (SD)*	*M (SD)*
Performance motivated	3.75 (3.68)	8.52 (6.07)	7.02 (5.32)	7.96 (5.86)	7.39 (6.22)	13.11 (8.91)
Performance not motivated	3.92 (3.87)	9.93 (6.78)	6.49 (5.30)	8.69 (7.10)	10.04 (9.06)	13.11 (8.09)
Performance with accuracy information	3.44 (3.07)	8.55 (6.3)	6.54 (5.08)	7.73 (6.47)	8.38 (7.06)	13.11 (8.436)
Performance without accuracy information	4.28 (4.40)	9.96 (6.52)	7.01 (5.56)	8.98 (6.50)	9.05 (8.67)	13.12 (13.12)

Notes. Low-quality recommendations were provided in T3 and T6. $N = 337$.

Table 4
Mixed ANOVA with high/low motivation and accuracy information as between factors; trials (six trials) as within factor and interaction effects.

	DFn	DFd	F	p	η^2
Motivation	1	333	4.43	0.036*	0.003
Accuracy information	1	333	4.50	0.035*	0.003
Trials	5	1665	77.86	0.000*	0.152
Motivation: Accuracy information	1	333	2.93	0.088	0.002
Motivation: Trials	5	1665	2.72	0.019*	0.006
Accuracy information: Trials	5	1665	0.51	0.766	0.001
Motivation: Accuracy information: Trials	5	1665	0.34	0.890	0.001

Notes. $N = 337$. *$p < .05$.

research suggested that self-reported trust may not reliably predict trusting behavior (Lai & Tan, 2019), we also included deviation from the recommendation as a behavioral indicator of trust. Indeed, there were significant differences between the high- and low-motivation conditions, with motivated participants deviating less from the AI recommendation, with $F(1,333) = 4.47$, $p < .05$, $\eta^2 = 0.013$. A mixed ANOVA revealed that the quality of the AI recommendation moderated the effect of motivation on deviation from recommendation, with $F(1,333) = 4.45$, $p < .05$, $\eta^2 = 0.006$. The motivated participants deviated less from the recommendation when the recommendation was of high quality (see Table 2). In summary, motivated participants (despite not reporting significantly higher trust) showed more trusting behavior in the AI system's recommendation than people in the low-motivation conditions, especially if the recommendation was of high quality (see Table 2).

4.3.2. Accuracy information

H2 stated that *accuracy information positively affects performance of human-AI collaborations in decision tasks with explanations.* In line with this hypothesis, providing accuracy information resulted in a significantly better performance, with $F(1,333) = 4.50$, $p <. 01$, $\eta^2 = 0.013$. To investigate this result, we examined in what rounds participants showed stronger performance. The results indicated that there was no interaction effect between quality of recommendation and the effect of accuracy information on performance, $F(1,333) = 1.06$, $p = .30$. The improvement of participants' performance was similar for high- and low-quality recommendations.

In RQ2, we wanted to explore *how accuracy information in human-AI collaborations in decision tasks with explanations influences trust towards the AI system.* The ANOVA showed no significant effect of accuracy information on self-reported trust $F(1,335) = 3.84$, $p = .05$. However, accuracy information significantly affected trusting behavior (deviation from AI recommendation), but only in interaction with the quality of the AI recommendation, with $F(1,333) = 8.28$, $p < .01$, $\eta^2 = 0.009$. Participants given accuracy information deviated more from low-quality recommendations and less from high-quality ones (see Table 2). In

summary, although accuracy information did not affect self-reported trust, it did increase trusting behavior towards high-quality recommendations and decrease trusting behavior towards low-quality recommendations.

Finally, in H3 we proposed that *we expect motivation and accuracy information to have independent effects on performance of human-AI collaborations in decision tasks with explanations, so that the combination of both factors should lead to a better performance than either factor alone.* The combination of motivation and accuracy information did not lead to a significantly higher performance than each condition by itself, with the interaction effect resulting in $F(1,333) = 2.93$, $p = .08$, thus not supporting our hypothesis.

5. Discussion

The present study investigated the influence of motivation and accuracy information on the joint decision-making performance of HAI collaboration, as well as the impact of these factors on human trust in AI recommendations with explanations. The results indicate that both motivation and accuracy information can improve performance in an experimental joint decision-making task. We also examined how both factors influence trust and found that while none of the factors influenced self-reported trust, both influenced trusting behavior (deviation from the AI system's recommendations). Motivated participants improved not by detecting errors of the AI system, but rather through better recognizing and more strongly trusting the high-quality recommendations. Accuracy information increased trusting behavior toward high-quality and decreased trusting behavior towards low-quality recommendations. Contrary to our expectations, the positive influence of motivation on performance was not moderated by need for cognition, and we also found no incremental value of the combination of high motivation and accuracy information above the positive effects on performance of each condition alone.

5.1. Theoretical and practical implications

We expected participants in the high-motivation condition to perform better in the performance task, as they would exert extra cognitive effort, thus they would be able to assess recommendations and explanations deliberately and use them more adequately. We confirmed the improvement in performance, but the process was different than expected. Rather than mainly reducing over-trust, as suggested by prior research (Chromik et al., 2021; Eiband et al., 2019), extrinsically motivated participants more strongly trusted in the system's recommendations (and deviated from them less) when the system provided high-quality outputs. Therefore, the improvements on performance may have originated from participants initially not trusting the system and then gaining trust as they were motivated to elaborate the recommendations and explanations. This emphasizes a point of Zerilli et al. (2022), in that the sole focus on either over-trust or under-trust is insufficient, since both play an important role in the (mis)calibration of trust in HAI collaboration.

In contrast to our expectations and previous research (Buçinca et al., 2021), we also did not find a moderating effect of need for cognition on the influence of motivation. This may be due to the high need for cognition in our sample, with over 75% of all participants scoring on the upper half of the scale. Since we employed a non-commercial research panel, consisting exclusively of members who voluntarily participate in surveys, it is plausible that the intrinsic motivation to exert cognitive effort of the members of this panel is comparably high. This limited variance regarding need for cognition in our sample makes the detection of a moderation effect less likely. Additionally, with a more diverse sample, we may even expect stronger effects of a motivation manipulation, since high intrinsic motivation to exert cognitive effort may have introduced a ceiling effect, with many participants already highly motivated prior to the manipulation.

The results confirm our expectations of improved performance when accuracy information is provided. Unlike motivation, the positive effects of accuracy information on performance do not seem dependent on the quality of the system's recommendation. Participants seemed to benefit from the accuracy information by being better at recognizing both high-quality recommendations and low-quality recommendations. This is also reflected in trusting behavior: Participants provided with accuracy information were more likely to deviate from low-quality recommendations and less likely to deviate from high-quality recommendations.

Although motivation increased performance with high-quality recommendations, and accuracy information increased performance for both recommendations, the combination of high motivation and accuracy information did not yield significant incremental effects on performance over each effect alone. One possible explanation is that the positive effect of motivation may already have been covered by the effect of accuracy information. Participants may not benefit from the additional motivation to identify good recommendations if the accuracy information has already identified the good recommendations for them. If this were the case, one hypothesis for future research could be that incremental effects of combining both factors only occur when the accuracy information is not perfectly reliable. In this case, people might benefit from the necessary motivation not to blindly rely on the accuracy information to warn them of potential errors.

Ultimately, our results suggest that under the right circumstances, people can make better use of AI system recommendations and explanations. One beneficial factor is the motivation to perform well, which allows for better trust calibration and, thus, better use of high-quality AI recommendations and explanations. The other beneficial factor is to facilitate trust calibration through providing accuracy information. The additional information indicating which recommendations are of higher quality (e.g., which is similar to many cases of the training data of the AI system) than others (e.g., edge cases) allows the human decision-maker to more adequately trust the system. The mechanisms of motivation and accuracy information may have been similar enough to be interchangeable in their effect on performance. This indicates that motivated engagement with the recommendations and explanations allows for a similar level of performance as when external information about the quality of recommendations is available.

To translate our findings into practice, it is important to consider that there are multiple ethical and practical considerations to be made. Baum et al. (2022) described the importance of closing "responsibility gaps", ensuring that, especially in high-stake decisions that affect humans directly, clarity is needed about who bears the responsibility for faulty decisions. The authors propose the necessity of humans "in the loop" who have access to the reasons of an AI system's recommendation, the ability to overrule the AI system, and the adequate expertise to make such decisions. The findings of our study address the subject of accuracy information and high motivation as two additional facets to provide humans with the ability to assess and use AI recommendations appropriately. The ethical confines of commercial AI use are currently discussed and prospectively regulated by upcoming laws, such as the EU AI Act. The implementation of AI-tasks could also benefit from more practical considerations. The balance of importance/criticality of a decision, responsibility for the outcome, and the desired precision and speed of task completion are important determinants of the adequate level of autonomy and transparency of an AI system.

5.2. Limitations and further research

It is important to interpret the conclusions of this study with some limitations in mind. The measures of self-reported trust and self-reported motivation did not change according to our expectations. However, the complementary behavioral measure (deviation from recommendation as a measure of trust; better performance as an indicator of motivation) did change based on the condition. Reasons for this pattern might be either the participants' limited ability for introspection or the timing of the

measure. We measured self-reported trust and motivation only in the very end of all performance trials. Initial differences in motivation and trust may have decreased through the course of the experiment. Future research might employ continuous motivation and trust measures to better monitor the trajectories within trials, especially after errors of the AI system are identified, and compare these measures with behavioral indicators of these constructs (Lai & Tan, 2019).

More research is needed on the mechanism by which motivation and accuracy information improve HAI decision-making. For example, based on our reasoning and findings of prior research, we expected that motivated people would elaborate more on the explanations of the system. However, it cannot be ruled out that the participants would have improved similarly if only provided with a recommendation, without an explanation. While our results identify a practical way to improve HAI collaboration, future research should further explore the complex mechanism behind these improvements.

It may also be relevant to acknowledge that our sample had little experience with both AI and personnel selection. We designed the performance task to not depend on prior knowledge in either field, but this may have influenced trust in participants' own abilities to perform the task and also trust in the system (Hoff & Bashir, 2015). However, since our sample included a diverse age range and participants were recruited via an online survey panel, it is likely to assume that they may be representative of the general working population. Because intelligent systems have only recently become popular in work environments, experience may still be scarce in general. Thus, it would be interesting to see how emerging experience with AI systems affects future HAI decision-making.

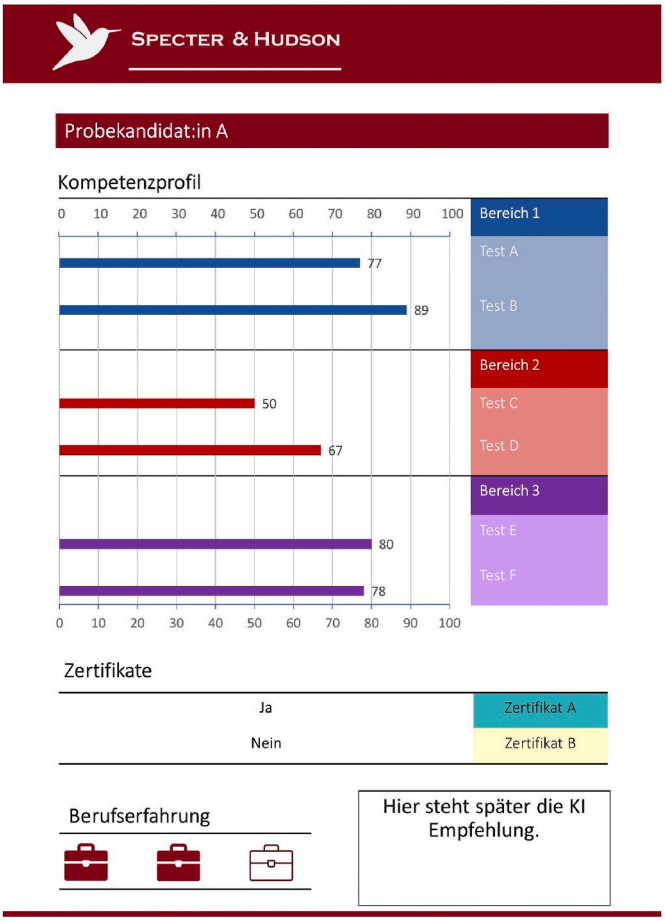
Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

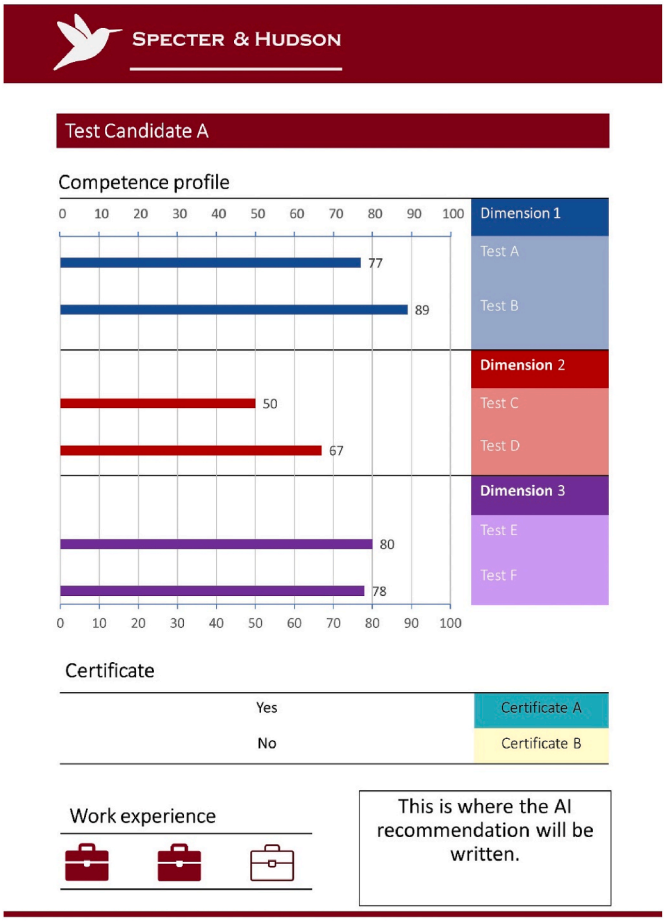
Appendix A. Materials

Appendix A.1. Applicant Profiles German Original and English Translation

German



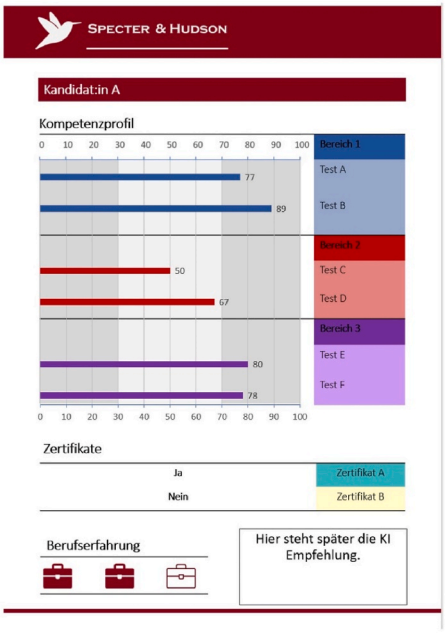
English



Appendix A.2. Evaluation Rules

Rechts sehen Sie ein Bewerberprofil.

- Ganz oben sind die Kompetenzbereiche 1, 2 und 3 angezeigt, die sich durch jeweils zwei Testwerte zusammensetzen.
- Die beiden Werte eines Kompetenzbereichs sind immer gleich wichtig. Bereich 1 ist dabei der wichtigste, Bereich 2 am unwichtigsten und Bereich 3 etwas weniger wichtig als Bereich 1.
- Darunter ist einsehbar ob Zertifikat A und B sichtbar sind.
- Wenn Zertifikat A fehlt, zieht das die Wertung deutlich runter.
- Das Fehlen von Zertifikat B beeinflusst die Wertung nur negativ wenn kaum Berufserfahrung vorhanden ist
- Die Berufserfahrung wird mit Aktentaschen dargestellt, mit minimal einer und maximal drei ausgefüllten Aktentaschen.
- Eine Berufserfahrung von zwei oder drei Aktentaschen macht nur einen kleinen Unterschied.



English Translation

On the right you can see an applicant profile. At the top, competency areas 1, 2, and 3 are displayed, which are composed of two test values each.

The two values of a competency area are always equally important. Area 1 is the most important, area 2 is the least important, and area 2 is slightly less important than area 1.

Below, it is possible to see whether certificates A and B are present.

If certificate A is missing, the rating is significantly lower.

The absence of certificate B only negatively affects the rating if there is little work experience.

Work experience is represented by briefcases, with a minimum of one and a maximum of three filled briefcases.

Having an experience of two or three briefcases only makes a small difference.

Appendix A.3. Performance Formula

```
def calculate(self):
    Tests = (((self.A + self.B)/2) *3) + (((self.C + self.D)/2) *0.7) + (((self.E + self.F)/2) *2.3)
    Tests = Tests /6

    if self.ZA == False:
        Tests == Tests *0.75
    if self.ZB == False & self.BE ==1 :
        Tests = Tests *0.8
    if self.BE == 2:
        Tests = Tests*0.95

    Tests = round(Tests,0)
    return(Tests)
```

Explanation: Each test dimension consists of two tests, and the scores of these tests are averaged. Then a score between 1 and 100 is calculated with dimension one counting in with a factor of 3, dimension two with a factor of 0.7, and dimension three with a factor of 2.3. Then, when the work experience is 1, 25% of the resulting score is subtracted if certificate A is missing, by 20% when certificate B is missing. Finally, if the work experience is 2 rather than 3, then the value is reduced by 5%.

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